# A Hierarchical Structure-Guided Cross-Modal Calibration Network for SAR–Optical Cloud Removal

Zhicheng Ye, Yufeng Liu, Hu Li, and Juluduzi ShaShan

***Abstract*—Cloud contamination remains a fundamental obstacle for optical Earth observation, severely limiting the availability and temporal continuity of remote sensing imagery. Although synthetic aperture radar (SAR) provides complementary all-weather structural information, existing SAR–optical cloud-removal methods remain challenged by three coupled issues: heterogeneous microwave-scattering and optical-reflectance characteristics complicate cross-modal feature integration; cloud occlusions vary substantially in spatial extent and therefore require adaptive contextual support; and reconstruction must preserve global scene consistency and local spectral–textural details without introducing excessive computational redundancy. To address these challenges, we propose a hierarchical structure-guided cross-modal calibration network (HySADNet) for SAR-assisted optical cloud removal. HySADNet adopts an asymmetric dual-stream architecture in which the optical pathway serves as the primary reconstruction stream, while the SAR pathway provides cloud-independent structural guidance. A cross-modal synergistic calibration and fusion module performs reciprocal feature recalibration and reliability-weighted fusion between the two modalities, while a dynamic kernel fusion (DKF) module adaptively aggregates multiscale contextual information. In addition, a structure-aware hybrid Transformer block (SAHTB) combines window-based self-attention with convolutional local priors to jointly model long-range dependencies and fine-grained structures. Experiments on SEN12MS-CR and LuojiaSET-OSFCR demonstrate the effectiveness of the proposed method. Compared with the corresponding second-best results, HySADNet improves peak signal-to-noise ratio (PSNR) by 0.2608 dB on SEN12MS-CR and by 0.7050 dB on LuojiaSET-OSFCR; it also reduces spectral angle mapper (SAM) by 0.1612 and 0.2947, respectively. Downstream semantic-segmentation evaluation further indicates that the reconstructed imagery preserves useful structural and spectral information for the evaluated high-level vision task.**

***Index Terms*—Cloud and shadow removal, deep learning, remote sensing, synthetic aperture radar (SAR)–optical data fusion.**

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## I. Introduction

OPTICAL remote sensing imagery, characterized by intuitive geometric structures and rich spectral textures, has become a fundamental data source for global land-cover classification, environmental monitoring, and disaster assessment [1], [2], [3]. However, due to the complexity of atmospheric conditions, cloud occlusion remains an inevitable obstacle in optical Earth observation [4]. A considerable portion of the Earth's surface is persistently covered by clouds, leading to severe loss of critical surface information during remote sensing acquisition [5], [6]. Consequently, addressing cloud contamination is of great scientific significance and practical value for improving the temporal continuity and spatial availability of optical remote sensing data [7].

To overcome the limitations imposed by cloud occlusion, existing cloud removal techniques have mainly evolved along three paradigms: single-image-based restoration [8], multitemporal reconstruction [9], and multimodal data fusion [10]. Single-image methods typically rely on spatial interpolation, which is often restricted to thin cloud removal or small-area restoration and struggles to cope with large-scale information loss. Multitemporal approaches exploit complementary information from time series observations; however, they are highly susceptible to temporal inconsistencies and hallucinated textures when rapid land surface changes occur, such as flooding events or vegetation phenological variations.

On the contrary, synthetic aperture radar (SAR), benefiting from its active microwave imaging capability that penetrates clouds and fog, can provide stable geometric structural information of the Earth's surface under all-weather and all-day conditions [11], [12]. As a result, SAR has been widely recognized as the most promising auxiliary data source for optical image cloud removal tasks [13], [14].

Despite the substantial progress achieved by SAR-assisted optical declouding methods, realizing high-fidelity cross-modal reconstruction still faces two critical technical challenges. First, SAR images are formed based on microwave scattering mechanisms and primarily reflect the dielectric properties and surface roughness of objects, whereas optical images record spectral reflectance characteristics [15]. This nonlinear radiometric discrepancy, coupled with the inherent geometric distortions and speckle noise in SAR imagery, makes simple feature concatenation or shallow fusion strategies insufficient for effectively bridging the modality gap

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or establishing accurate structure–spectral mapping relationships [16], [17]. Existing methods often fail to adequately disentangle modality-specific private features from shared representations, leading to reconstruction results that either excessively retain SAR speckle noise in texture or suffer from severe spectral distortion in optical appearance. Although recent studies have attempted to separate modal representations through disentangled learning strategies [18], accurately aligning heterogeneous features in complex scenes remains an unresolved challenge.

Second, cloud formations in real-world scenarios exhibit highly diverse and complex morphologies, ranging from fragmented thin clouds to extensive thick cloud clusters spanning several kilometers. Conventional convolutional neural network (CNN)-based methods are constrained by fixed local receptive fields, making it difficult to capture long-range contextual dependencies when dealing with large-scale thick cloud occlusion, often resulting in blurred or blocky artifacts in the reconstructed regions [19]. Although Xie et al. [20] introduced a stereo attention mechanism to enhance global modeling capability, and Chouhan et al. [21] further explored the application of diffusion models for cloud removal, the high computational overhead of pure Transformer architectures and the low inference efficiency of diffusion models significantly hinder their applicability to large-scale, real-time remote sensing processing tasks.

Despite significant progress in multimodal remote sensing restoration, three critical challenges remain insufficiently addressed in current SAR–optical fusion-based cloud removal frameworks: cross-modal heterogeneity, where the stark differences in imaging mechanisms and statistical distributions between microwave scattering (SAR) and spectral reflectance (optical) hinder stable feature correspondence; multiscale cloud structures, where diverse spatial patterns ranging from thin fragmented clouds to extensive dense masses necessitate adaptive receptive field modeling to prevent information loss or blurring; and the joint modeling of global structure and local texture, where conventional architectures struggle to maintain global scene consistency while simultaneously recovering fine-grained textures under heavy computational constraints.

In light of these limitations in modality heterogeneity mitigation and multiscale feature aggregation, this article proposes the hierarchical structure-guided cross-modal calibration network (HySADNet). The core motivation of HySADNet lies in breaking through the performance bottleneck of traditional symmetric architectures by constructing a deeply collaborative mechanism in which structural priors guide spectral calibration while incorporating a multiscale adaptive perception strategy tailored to cloud distribution. By utilizing an asymmetric dual-stream design—deploying a hierarchical Transformer for the optical stream to capture long-range dependencies and a lightweight CNN for the SAR stream to extract geometric anchors—the model ensures that optical reconstruction is primary while SAR features provide stable structural guidance. This approach enables high-fidelity reconstruction of optical imagery even under complex surface conditions and extreme thick cloud occlusion. The main contributions of this work are summarized as follows.

1) We formulate SAR–optical cloud removal as an asymmetric reconstruction problem according to the unequal roles of the two modalities. Optical imagery serves as the primary spectral reconstruction target, whereas SAR imagery provides cloud-independent structural guidance. Based on this formulation, we develop a hierarchical asymmetric dual-stream architecture that allocates different representation capacities to the optical and SAR pathways, thereby avoiding unnecessary duplication between modalities with substantially different information characteristics and task roles.
2) To address heterogeneous cross-modal responses and scale-varying cloud occlusion, we introduce a cross-modal synergistic calibration and fusion module (CSCM) and a dynamic kernel fusion (DKF) module. CSCM performs reciprocal cross-modal feature calibration and reliability-weighted fusion, allowing SAR-derived structural cues to recalibrate cloud-contaminated optical representations without implying explicit geometric or distribution alignment. DKF dynamically aggregates multiscale contextual information so that the effective contextual support can adapt to cloud occlusions with different spatial extents.
3) To jointly model global scene structure and local spectral/textural details while controlling model complexity, we develop a structure-aware hybrid Transformer block (SAHTB) within the reconstruction pathway. SAHTB combines window-based self-attention for long-range dependence modeling with a convolutional spatial-structure branch for local detail representation. Together with the asymmetric encoder design, this global–local representation strategy aims to retain strong reconstruction capability while limiting redundant model capacity.
4) Extensive experiments on SEN12MS-CR and LuojiaSET-OSFCR demonstrate competitive reconstruction performance under the evaluated paired SAR–optical benchmark settings. Controlled replacement and computational analyses further characterize the contribution and accuracy–complexity tradeoffs of the proposed architecture.

## II. Related Work

### A. SAR–Optical Heterogeneous Feature Fusion for Cloud Removal

In multimodal declouding tasks, the central challenge lies in establishing an effective mapping between microwave-derived geometric information and optical spectral representations. Early studies predominantly adopted implicit fusion strategies. For instance, SAR-opt-cGAN proposed by Grohnfeldt et al. [22] and DSen2-CR introduced by Meraner et al. [23] incorporated SAR features into the network through channel concatenation or elementwise addition. Although these approaches partially exploited the cloud-penetrating capability of SAR, they largely ignored the intrinsic distribution discrepancy between modalities, which often resulted in the direct transfer of SAR speckle noise into the reconstructed optical images.

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To enable more refined feature selection, attention mechanisms were subsequently introduced. Xu et al. [24] proposed the GLF-CR model, which employed a gated mechanism to reweight fused features in an attempt to suppress irrelevant information. Li et al. [25] developed the HS2P model, leveraging channel attention to enhance spectral fidelity. Tu et al. [26] further introduced a multimodal similarity attention mechanism, in which similarity maps between SAR and optical features were explicitly computed to guide texture restoration.

In recent years, to better cope with complex modality heterogeneity, research efforts have gradually shifted toward feature disentanglement and more sophisticated interaction mechanisms. Gu et al. [27] proposed the HPN-CR model, which designed heterogeneous parallel blocks to separately process SAR and optical modalities, emphasizing the necessity of modality-specific feature extraction. He et al. [18] advanced generative adversarial networks (GANs) by proposing the DADIGAN model, which employed dual-attention blocks for iterative feature disentanglement, effectively separating cloud-induced noise from underlying land-surface backgrounds. Anandakrishnan et al. [28] combine SAR guidance with a residual multiscale dilated network for reconstructing cloud-contaminated Sentinel-2 imagery, demonstrating the value of multiscale contextual modeling in multimodal cloud removal. Inspired by these studies, this work advances cross-modal fusion by introducing a synergistic calibration mechanism. Instead of simple concatenation, this approach allows SAR-derived structural information to explicitly calibrate optical spectral representations, facilitating reciprocal cross-modal feature recalibration and information integration.

### *B. From Local Long-Range Modeling to Dynamic Context Aggregation*

When dealing with large-scale thick cloud occlusion, the ability to effectively aggregate contextual information from cloud-free regions is critical to reconstruction quality. The inherent locality of conventional convolutional operations makes it difficult to exploit global contextual cues under such conditions, often leading to blurred textures or structural discontinuities in the reconstructed regions [29], [30], [31].

To overcome this limitation, Transformer-based architectures have been introduced into cloud removal tasks. TASANet proposed by Xie et al. [20] employed a bound-anchored stereo attention mechanism to establish explicit feature correspondences between reference and target images, significantly enhancing global structural recovery. Li et al. [32] proposed a dual-domain cross-modal interaction network that leveraged Transformers to perform global context modeling in deep feature spaces. Meanwhile, diffusion models have recently emerged as a promising direction due to their strong generative capabilities. The DGMR model proposed by Chouhan et al. [21] adopted a diffusion-guided masked reconstruction strategy, demonstrating remarkable performance under extreme occlusion conditions. Zou et al. [33] introduced the DiffCR model, which improved perceptual quality by incorporating a disentangled conditional encoder and a temporal–conditional fusion module. Liu et al. [34] further developed the EMRDM model, which reformulated the denoising process using a mean-regression stochastic differential equation, effectively exploiting residual information in cloudy images to mitigate content hallucination under severe occlusion.

However, these advanced methods are often accompanied by substantial computational costs: Transformer-based processing can become expensive as spatial resolution increases, while diffusion models require iterative sampling during inference. Motivated by these limitations, we combine convolutional local priors with window-based self-attention rather than relying exclusively on either CNN or Transformer processing. The objective is to retain long-range representation capability while controlling unnecessary computational redundancy.

## III. Method

This section presents the detailed architecture of the proposed HySADNet. By fully exploiting the complementary characteristics of Sentinel-1 SAR data and Sentinel-2 multispectral optical imagery, HySADNet is capable of reconstructing cloud-free optical images from cloud-contaminated observations. The overall objective of the proposed method is to learn a mapping function $\mathcal{M}$, which reconstructs a clean optical image $I_{clean}$ from a cloud-contaminated optical image $I_{opt}$, under the guidance of a spatially coregistered auxiliary SAR image $I_{sar}$.

### *A. Overall Structure*

HySADNet adopts a hierarchical asymmetric encoder–decoder architecture according to the different roles of SAR and optical imagery in the reconstruction task. The optical observation is the spectral reconstruction target, whereas the SAR observation acts as auxiliary structural guidance. Moreover, the optical input contains 13 spectral bands, while the SAR input contains two polarization channels. These differences motivate the use of modality-specific representation pathways rather than assigning identical processing capacity to both streams.

As illustrated in Fig. 1, let $I_{opt} \in \mathbb{R}^{H\times W\times C_{opt}}$ and $I_{sar} \in \mathbb{R}^{H\times W\times C_{sar}}$ denote the input optical and SAR images, respectively. The optical stream employs a hierarchical Transformer encoder to capture spatial–spectral interactions and long-range contextual dependencies, whereas the SAR stream uses a lightweight CNN encoder to extract geometric and local structural cues. This configuration is designed to concentrate representational capacity on the primary optical reconstruction path, whilst limiting redundant computations in the auxiliary SAR path. Importantly, we do not make the a priori assumption that asymmetric processing is superior to symmetric processing in all cases; in subsequent ablation experiments, we evaluate this design choice against alternatives such as CNN–CNN and Transformer–Transformer.

Let $E^i_{opt}$ and $E^i_{sar}$ denote the encoding operations at the $i$th level ($i \in \{1,\ldots,4\}$) of the optical and SAR encoders, respectively. The feature extraction process can be formulated as

$$X^i_{opt} = E^i_{opt}\left(X^{i-1}_{opt}\right), \quad X^i_{sar} = E^i_{sar}\left(X^{i-1}_{sar}\right) \tag{1}$$

where $X^0$ represents the initial embedding of the raw input images. The deep features $X^i_{opt}$ and $X^i_{sar}$ are subsequently fed

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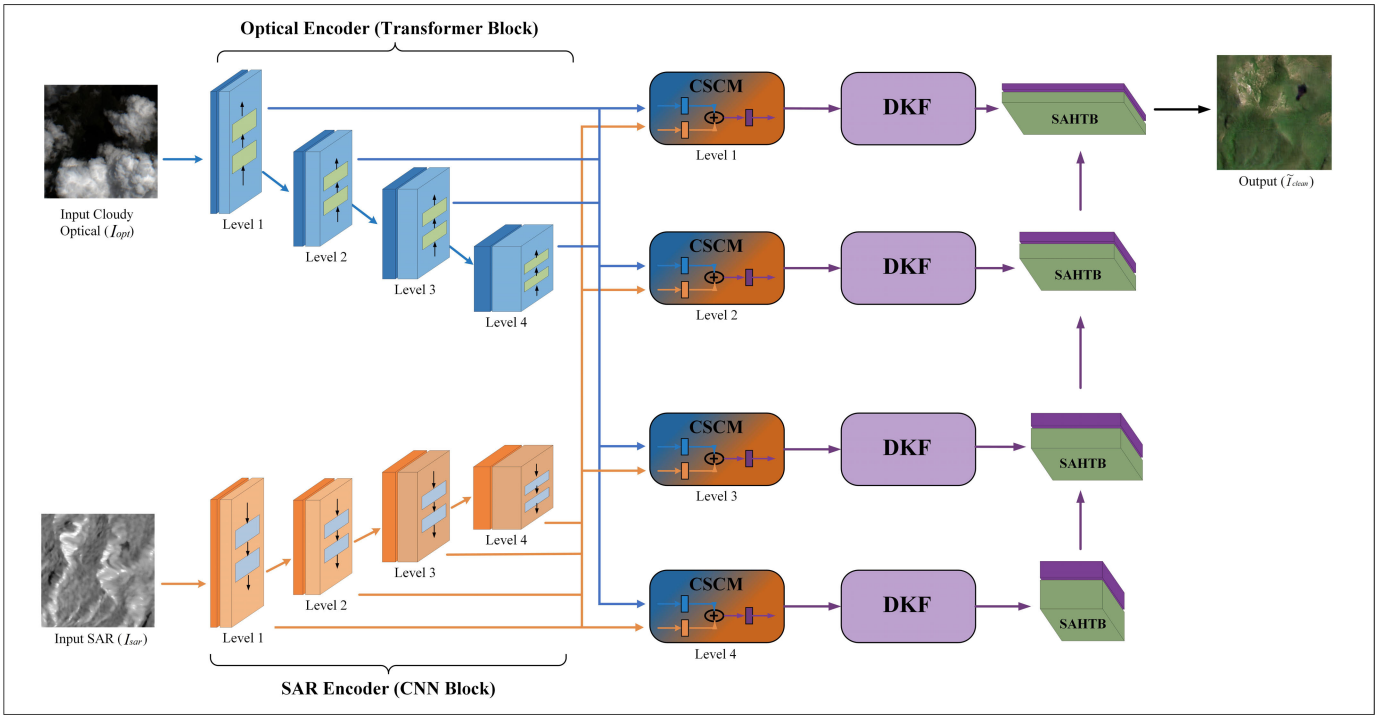



Fig. 1. Overall architecture of the proposed HySADNet framework.

into the CSCM for cross-modal integration. At each hierarchical level, CSCM leverages SAR-derived geometric structural information to perform deep collaborative calibration and fusion of cloud-contaminated optical features. The resulting calibrated features are then passed to the DKF module.

In particular, at the $l$th decoding stage, the DKF module receives the calibrated fusion features from CSCM and dynamically aggregates multiscale contextual information via an adaptive receptive field mechanism

$$F^l_{fused} = \mathrm{DKF}^l\left(\mathrm{CSCM}\left(E^l_{opt}, E^l_{sar}\right)\right). \tag{2}$$

The fused features are subsequently integrated into the decoder backbone. The $l$th decoding unit takes the upsampled features $D^{l+1}$ from the deeper layer and concatenates them with the current fusion features $F^l_{fused}$. The concatenated features are then refined by a SAHTB, formulated as

$$D^l = \mathrm{SAHTB}^l\left(\mathrm{Concat}\left(\mathrm{Upsample}\left(D^{l+1}\right), F^l_{fused}\right)\right). \tag{3}$$

Finally, the cloud-free optical image $\tilde{I}_{clean}$ is generated at the end of the decoder through a terminal convolutional layer.

### *B. Cross-Modal Synergistic Calibration and Fusion Module(CSCM)*

Direct fusion of SAR and optical features is challenging because the two modalities exhibit substantially different radiometric characteristics and noise patterns. CSCM is therefore designed as a reciprocal feature-calibration and reliability-weighted fusion mechanism rather than as a geometric alignment module. Given coregistered inputs spatially, its objective is to allow descriptors derived from one modality to recalibrate the feature response of the other modality and to adaptively control their contributions to the fused latent representation. The overall structure of CSCM is illustrated in Fig. 2.

Let $F_{opt}, F_{sar} \in \mathbb{R}^{C\times H'\times W'}$ denote the feature tensors input to CSCM. First, $1 \times 1$ convolutions are applied to align their channel dimensions, yielding transformed optical and SAR features. Meanwhile, both global average pooling ($\mathcal{G}_{avg}$) and global max pooling ($\mathcal{G}_{max}$) are employed to generate comprehensive modality descriptors

$$\begin{aligned} z_m &= W_{\text{shared}}\left(\delta\left(W_{\text{sq}}\left(\mathcal{G}_{\text{avg}}\left(F_m\right) + \mathcal{G}_{\text{max}}\left(F_m\right)\right)\right)\right) \\ m &\in \{\text{opt}, \text{sar}\} \end{aligned} \tag{4}$$

where $W_{sq}$ and $W_{shared}$ constitute a shared squeeze-and-excitation MLP. Subsequently, cross-modal attention maps are derived via a sigmoid activation function $\sigma$

$$\mathcal{A}_{sar\to opt} = \sigma\left(z_{sar}\right), \quad \mathcal{A}_{opt\to sar} = \sigma\left(z_{opt}\right). \tag{5}$$

These attention descriptors are used to reciprocally modulate the corresponding feature streams

$$\hat{F}_{opt} = F_{opt} \odot \mathcal{A}_{sar\to opt} + F_{opt} \tag{6}$$

$$\hat{F}_{sar} = F_{sar} \odot \mathcal{A}_{opt\to sar} + F_{sar}. \tag{7}$$

To enhance robustness against modality-specific noise, a dynamic gating mechanism is introduced to control the final fusion. The calibrated features are concatenated and processed by a fusion subnetwork $\mathcal{F}_{fuse}$ to generate a spatial confidence map $M$

$$M = \sigma\left(\mathcal{F}_{fuse}\left(\left[\hat{F}_{opt}, \hat{F}_{sar}\right]\right)\right). \tag{8}$$

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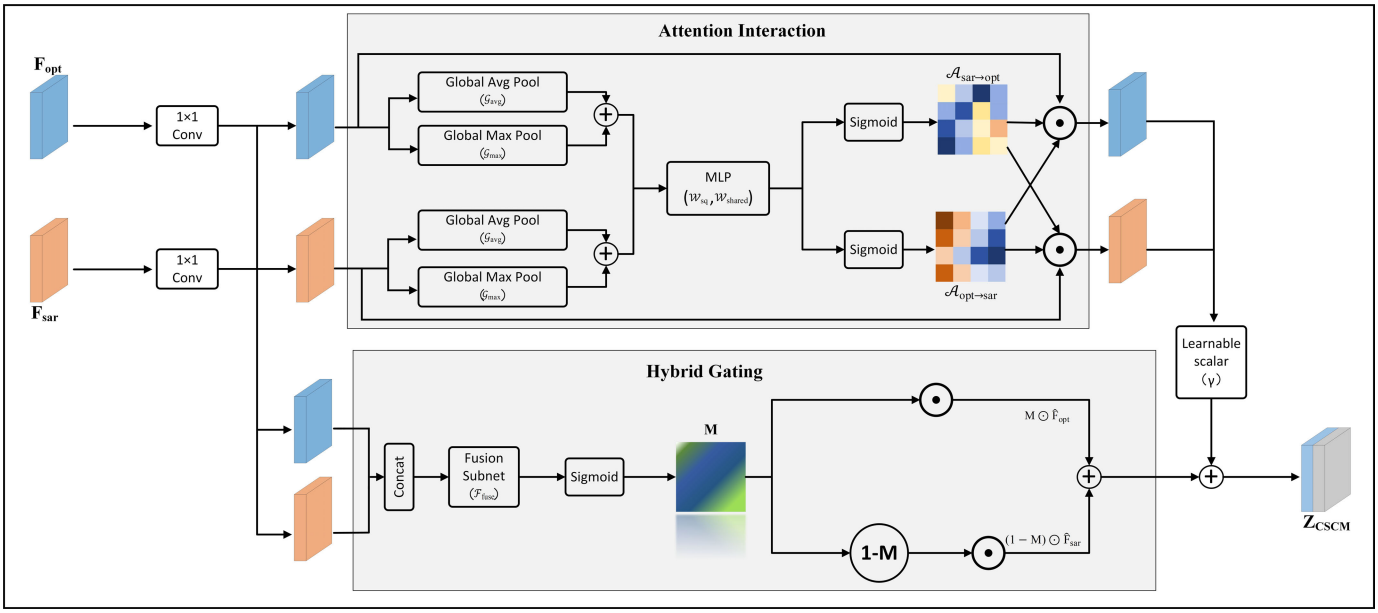



Fig. 2. Structure of the CSCM.

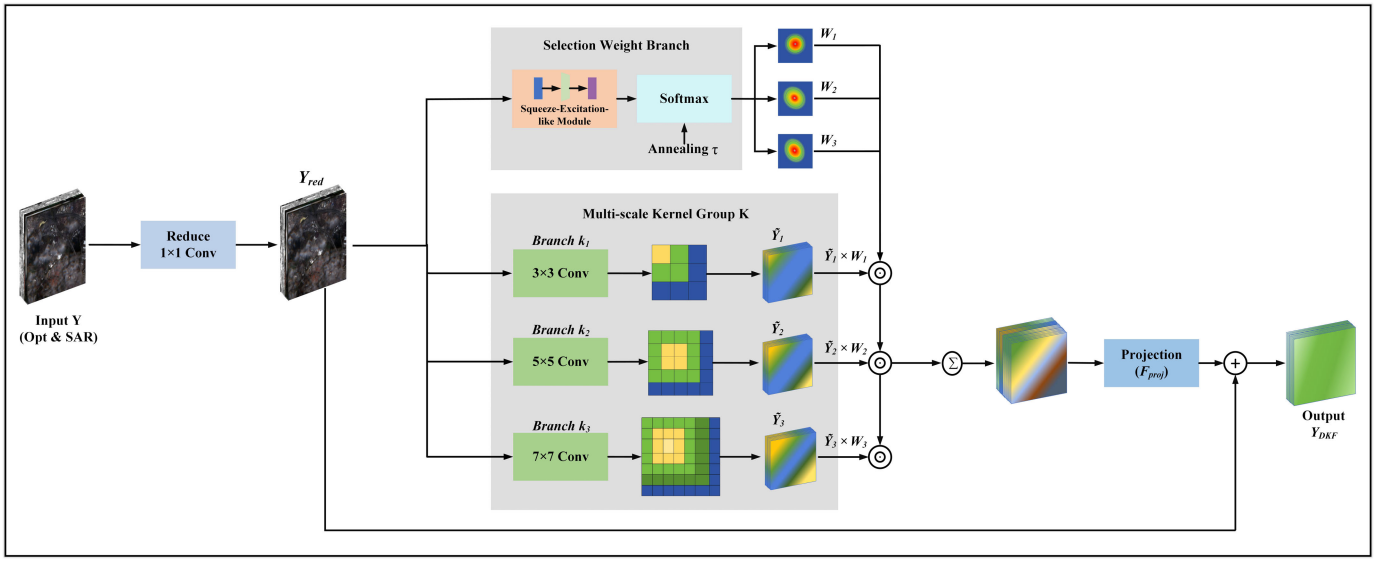



Fig. 3. Structure of the DKF module.

The final fused latent representation $Z_{CSCM}$ is obtained via a weighted averaging scheme, further refined by a global residual connection parameterized by a learnable scalar $\gamma$

$$Z_{CSCM} = \left(M \odot \hat{F}_{opt} + (1 - M) \odot \hat{F}_{sar}\right) + \gamma \cdot \left(\hat{F}_{opt} + \hat{F}_{sar}\right). \quad (9)$$

This design enables the fused representation to adaptively emphasize the most informative modality at each spatial location.

It should be emphasized that this operation constitutes latent feature calibration rather than geometric registration or explicit distribution matching. The cross-modal descriptors in (5)–(7) modulate feature responses, whereas the spatial confidence map in (8) controls the relative modality contribution at each location. CSCM therefore does not estimate spatial displacement between SAR and optical observations and does not mathematically guarantee equality between their feature distributions.

## C. DKF Module

Although selective-kernel and multiscale convolutional strategies are established approaches for handling scale variation, cloud occlusions exhibit large spatial variability that motivates adaptive contextual aggregation. DKF constructs a multiscale kernel set and uses a learnable selection mechanism to adjust branch weights according to the input features. As training progresses, the weighting mechanism allows the network to vary the relative contribution of different receptive-field scales, thereby providing adaptive contextual support for cloud regions with different spatial extents. An overview of DKF is shown in Fig. 3.

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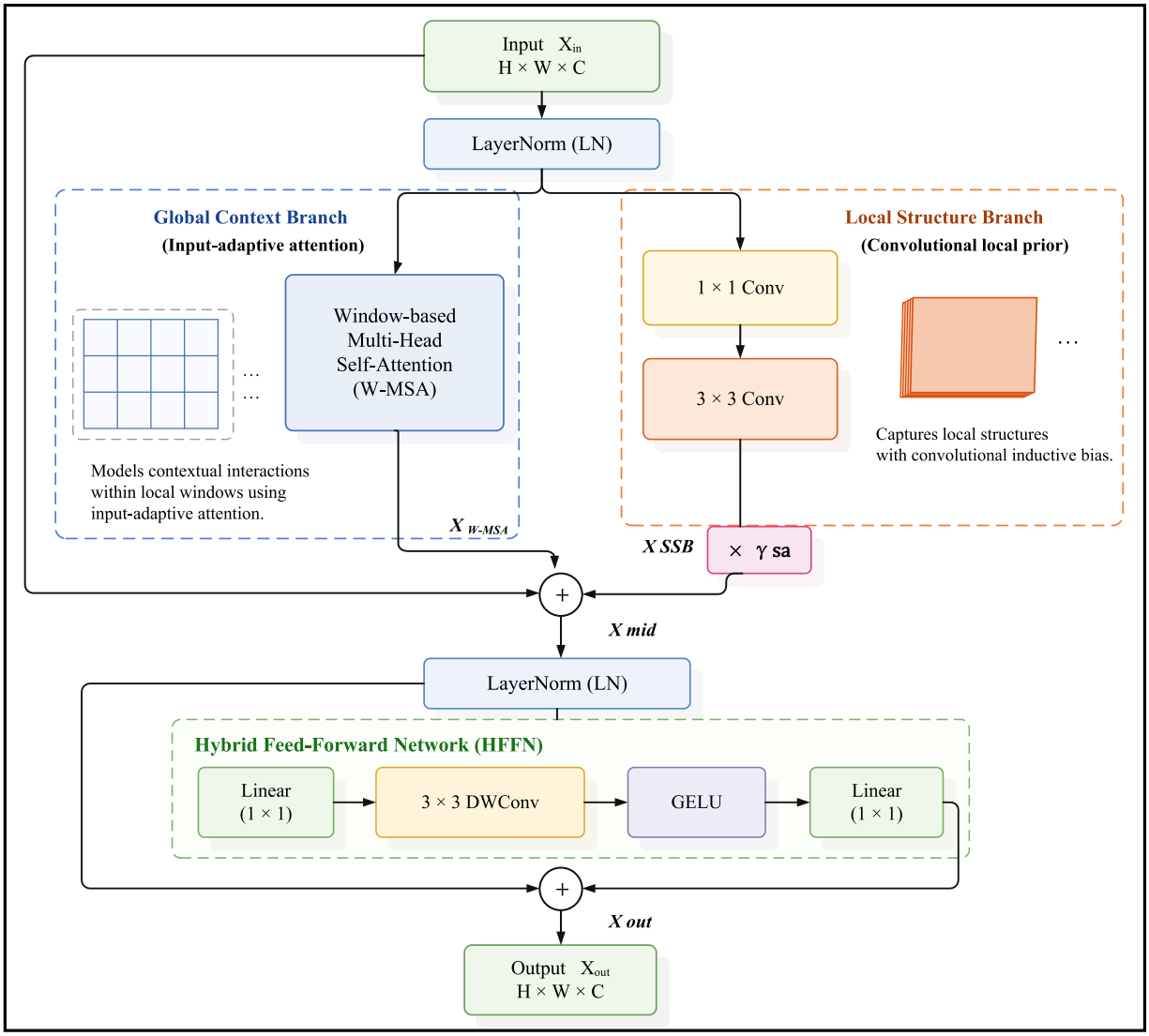



Fig. 4. Structure of the SAHTB.

In particular, DKF receives concatenated optical and SAR features from a given encoder level, denoted as $Y \in \mathbb{R}^{C_{in} \times H \times W}$. A $1 \times 1$ convolution is first applied for channel reduction, producing $Y_{red} \in \mathbb{R}^{C \times H \times W}$. A multibranch kernel group $K = \{K_1, K_2, K_3\}$ is constructed, consisting of a $3 \times 3$ convolution (local texture), a $5 \times 5$ convolution (medium-range context), and a $7 \times 7$ convolution (long-range receptive field). The resulting feature maps are

$$\tilde{Y}_k = \mathrm{Conv}\left(Y_{red}; K_k\right), \quad k \in \{1, 2, 3\}. \tag{10}$$

To determine the optimal kernel contribution for each pixel, a temperature-controlled selection mechanism is employed. A squeeze-and-excitation-like module predicts selection scores $S \in \mathbb{R}^{3 \times H \times W}$ from $Y_{red}$. The spatial weight for the $k$th kernel, $W_k \in \mathbb{R}^{1 \times H \times W}$ is computed using a softmax function with temperature parameter $\tau$

$$W_k(i, j) = \frac{\exp\left(S_k(i, j)/\tau\right)}{\sum_{n=1}^{3} \exp\left(S_n(i, j)/\tau\right)}. \tag{11}$$

During training, $\tau$ is linearly annealed from a high value (encouraging cross-scale exploration) to a low value (sharpening the selection toward the optimal scale). In particular, $\tau$ is linearly decreased from an initial value of 2.0 to a terminal value of 0.5 throughout the entire training duration. This scheduling strategy facilitates a seamless transition from broad multiscale exploration in the early stages to precise, localized receptive field selection as the network converges. The final scale-aware output is computed as

$$Y_{DKF} = \mathcal{F}_{proj}\left(\sum_{k=1}^{3} W_k \odot \tilde{Y}_k\right) + Y_{red}. \tag{12}$$

Through DKF, the network adaptively enlarges its receptive field in heavily clouded regions to leverage contextual cues while constraining the receptive field in clear areas to preserve high-frequency details.

### D. Structure-Aware Hybrid Transformer Block

Window-based self-attention provides an effective mechanism for modeling nonlocal dependencies, whereas convolution introduces useful locality and translation-related inductive

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biases for low-level image restoration. We therefore employ SAHTB as a hybrid global–local representation block that combines an input-adaptive attention pathway with a convolutional spatial-structure pathway (Fig. 4). The purpose of this design is to complement long-range contextual modeling with local structural representation rather than to make the two operations algebraically equivalent.

Given an input feature $X_{in}$, layer normalization is first applied, $X_{norm} = \mathrm{LN}(X_{in})$. The primary pathway uses window-based multihead self-attention

$$X_{\text{W-MSA}} = \text{W-MSA}\left(X_{norm}\right) \tag{13}$$

while the spatial structure branch (SSB) employs sequential $1 \times 1$ and $3 \times 3$ convolutions

$$X_{SSB} = \mathrm{Conv}_{3\times3}\left(\mathrm{Conv}_{1\times1}\left(X_{norm}\right)\right). \tag{14}$$

The two complementary representations are combined with a learnable scaling parameter $\gamma_{sa}$ and a residual connection

$$X_{mid} = X_{\text{W-MSA}} + \gamma_{sa} X_{SSB} + X_{in}. \tag{15}$$

Feature refinement is subsequently performed by a hybrid feed-forward network (HFFN), in which depthwise convolution is inserted between channel projections to strengthen local interactions

$$\mathrm{HFFN}\left(x\right) = W_2\left(\delta\left(\mathrm{DWConv}_{3\times3}\left(W_1\left(x\right)\right)\right)\right) \tag{16}$$

$$X_{out} = \mathrm{HFFN}\left(\mathrm{LN}\left(X_{mid}\right)\right) + X_{mid}. \tag{17}$$

It is important to distinguish the two parallel operations mathematically. The attention matrix in W-MSA depends on the input through

$$\mathrm{Softmax}\left(QK^{\mathsf{T}}/\sqrt{d}\right) \tag{18}$$

and therefore cannot, in general, be exactly folded into a fixed convolutional kernel or static projection matrix. Although compatible static convolutional operations may be algebraically composed under appropriate conditions, such a composition does not eliminate or reproduce the input-dependent attention coefficients. We therefore retain W-MSA as a dynamic attention operation and describe SAHTB as a hybrid global–local block. Its computational characteristics are assessed empirically in the efficiency analysis rather than inferred from an exact reparameterization claim.

### E. Loss Function

The training objective combines pixelwise reconstruction, multiscale structural similarity, and angular consistency

$$\mathcal{L}_{\text{total}} = 0.2\,\mathcal{L}_{\text{SmoothL1}} + 0.8\left(1 - \text{MS-SSIM}\right) + 0.005\,\mathcal{L}_{\text{ang}}. \tag{19}$$

The smooth $L_1$ term is computed with mean reduction over the reconstructed and reference optical tensors. MS-structural similarity index (SSIM) is evaluated over all 13 optical channels with a data range of 1.

For the angular term, let $\hat{I}, I \in \mathbb{R}^{B\times C\times H\times W}$ denote the reconstructed and reference optical tensors, respectively, where $C = 13$. For the $b$th sample and the $c$th spectral band, the corresponding spatial maps are vectorized as $\hat{x}_{b,c}, x_{b,c} \in \mathbb{R}^{HW}$. Their angular discrepancy is defined as

$$\theta_{b,c} = \arccos\left(\frac{\hat{x}_{b,c}^{\mathsf{T}} x_{b,c}}{\|\hat{x}_{b,c}\|_2 \|x_{b,c}\|_2}\right)\frac{180}{\pi}. \tag{20}$$

The angular loss is obtained by averaging over all samples and spectral bands in the minibatch

$$\mathcal{L}_{\text{ang}} = \frac{1}{BC}\sum_{b=1}^{B}\sum_{c=1}^{C}\theta_{b,c}. \tag{21}$$

The angular consistency term is calculated directly from the normalized inner product; prior to applying the arccos operation, neither an additional denominator regularization term is introduced nor are the cosine values clipped. This term measures the directional consistency between the reconstructed and reference spatial responses of each spectral band and complements the pixel-wise and structural reconstruction losses.

## IV. Experiments

### A. Datasets

To evaluate the reconstruction performance of the proposed method under two paired SAR–optical benchmark settings, we use SEN12MS-CR and LuojiaSET-OSFCR.

*1) SEN12Ms-CR Dataset:* SEN12MS-CR contains 122 218 triplets of spatially coregistered Sentinel-1 SAR, cloudy Sentinel-2, and cloud-free Sentinel-2 observations acquired from globally distributed regions of interest (ROIs). Each ROI is divided into $256 \times 256$ image patches, and neighboring patches may spatially overlap as part of the original dataset construction. In our experiments, no random repartition was performed for SEN12MS-CR; we directly followed the dataset-provided official ROI/scene-level split. All patches belonging to the same ROI were retained within the same subset, so spatially overlapping neighboring patches from a common ROI do not occur across the training, validation, and test sets. The resulting partition contains 107 143 training samples, 7176 validation samples, and 7899 test samples.

*2) LuojiaSET-OSFCR Dataset:* LuojiaSET-OSFCR is another important dataset focused on cloud removal and reconstruction of optical satellite imagery. Similar to SEN12MS-CR, it provides paired Sentinel-1 SAR and Sentinel-2 optical images. However, LuojiaSET-OSFCR additionally offers pixel-level auxiliary annotations, including cloud masks and land-cover classification maps, allowing for more fine-grained performance evaluation. The dataset contains a total of 20 000 high-quality paired samples. The pregenerated image patches were randomly partitioned into 12 000 training samples, 4000 validation samples, and 4000 test samples at a ratio of 6:2:2 using a fixed random seed of 100. Each patch was assigned exclusively to one subset, and the image patches used in different subsets are spatially nonoverlapping. Therefore, no identical or directly overlapping image content is shared across the training, validation, and test sets. However, because the partition was performed at the patch level rather than by enforcing

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region- or source-scene-disjoint subsets, this split should not be interpreted as a strict geographic holdout.

### B. Experimental Setup

*1) Implementation Details:* The proposed model was implemented in PyTorch and trained on an NVIDIA GeForce RTX 5090-D v2 GPU. The AdamW optimizer was adopted with an initial learning rate of $1 \times 10^{-4}$, which was decayed by a factor of 0.8 every five epochs. The model was trained for 30 epochs with a batch size of 16. All baseline methods were locally reproduced and evaluated in the same experimental environment. To ensure a consistent comparison, all reproduced methods used the same dataset partitions, input preprocessing, and evaluation pipeline as HySADNet. Peak signal-to-noise ratio (PSNR), SSIM, spectral angle mapper (SAM), and mean absolute error (MAE) were computed using the same metric implementations for all methods, while the method-specific network architectures followed their corresponding published designs. For inference profiling, all models were evaluated on the same NVIDIA GeForce RTX 5090-D v2 GPU using $256 \times 256$ inputs, batch size 1, FP32 precision, ten warm-up iterations, and 30 CUDA-synchronized timed runs.

For input preprocessing, the Sentinel-2 images were directly read from the spatially harmonized 13-band patches provided by the benchmark dataset, in which all bands are represented on a common 10-m spatial grid. No additional bandwise resampling was performed in our data-loading pipeline. Sentinel-2 values were clipped to $[0, 10\,000]$ and scaled to $[0, 1]$ by division by 10 000. For Sentinel-1, the VV and VH channels were clipped to $[-25, 0]$ and $[-32.5, 0]$ dB, respectively, followed by independent channelwise normalization to $[0, 1]$.

*2) Evaluation Metrics:* Four quantitative metrics are used to evaluate reconstruction quality: PSNR, SSIM, SAM, and MAE. Before metric computation, the reconstructed optical images are clipped to $[0, 1]$. All metrics are computed between the reconstructed image and the corresponding cloud-free reference, and the reported values are averaged over the test set.

PSNR measures the pixel-level reconstruction error. Since the optical images are normalized to $[0, 1]$, the peak intensity is set to $MAX_I = 1$

$$\mathrm{PSNR} = 10 \log_{10} \left( \frac{1}{\mathrm{mse}} \right) \tag{22}$$

where

$$\mathrm{mse} = \frac{1}{CHW} \sum_{c=1}^{C} \sum_{i=1}^{H} \sum_{j=1}^{W} \left( I_{c,i,j}^{gt} - I_{c,i,j}^{rec} \right)^2 \tag{23}$$

with $C = 13$ for Sentinel-2 imagery.

SSIM evaluates local structural consistency between the reconstructed and reference images

$$\mathrm{SSIM}(x, y) = \frac{\left(2\mu_x\mu_y + C_1\right)\left(2\sigma_{xy} + C_2\right)}{\left(\mu_x^2 + \mu_y^2 + C_1\right)\left(\sigma_x^2 + \sigma_y^2 + C_2\right)} \tag{24}$$

where $\mu_x$ and $\mu_y$ denote local means, $\sigma_x^2$ and $\sigma_y^2$ denote local variances, and $\sigma_{xy}$ denotes the local covariance. SSIM is evaluated using an $11 \times 11$ Gaussian window with $\sigma = 1.5$, $C_1 = 0.01^2$, and $C_2 = 0.03^2$. The SSIM scores are computed over the 13 optical channels and averaged to obtain the final image-level score.

SAM evaluates spectral fidelity by measuring the angular difference between the reconstructed and reference spectral vectors at each spatial location. Let $s_{i,j}^{rec} \in \mathbb{R}^{13}$ and $s_{i,j}^{gt} \in \mathbb{R}^{13}$ denote the reconstructed and ground-truth spectral vectors at pixel $(i, j)$, respectively. The spectral angle is defined as

$$\theta_{i,j} = \arccos \left( \frac{\left(s_{i,j}^{gt}\right)^{\mathrm{T}} s_{i,j}^{rec}}{\|s_{i,j}^{gt}\|_2 \|s_{i,j}^{rec}\|_2} \right) \frac{180}{\pi}. \tag{25}$$

The SAM value of an image is obtained by averaging the pixelwise spectral angles

$$\mathrm{SAM} = \frac{1}{HW} \sum_{i=1}^{H} \sum_{j=1}^{W} \theta_{i,j}. \tag{26}$$

Therefore, the lower the SAM value, the better the preservation of the spectral shape. For SAM, the same numerical treatment as for the angular consistency term is adopted; that is, no additional denominator regularization or cosine truncation is introduced prior to the arccos operation.

MAE measures the average absolute reconstruction error over all spectral bands and spatial locations

$$\mathrm{MAE} = \frac{1}{CHW} \sum_{c=1}^{C} \sum_{i=1}^{H} \sum_{j=1}^{W} \left| I_{c,i,j}^{gt} - I_{c,i,j}^{rec} \right|. \tag{27}$$

Higher PSNR and SSIM values indicate better reconstruction quality, whereas lower SAM and MAE values indicate better spectral fidelity and lower reconstruction error, respectively.

### C. Experimental Results

To comprehensively evaluate the effectiveness of HySADNet, we conduct extensive comparisons with several mainstream cloud-removal methods. The experiments are performed on two benchmark datasets, SEN12MS-CR and LuojiaSET-OSFCR, and the performance is assessed through both quantitative metrics and qualitative visual analysis. For a consistent qualitative comparison, the same predetermined test samples are used for all methods, and no method-dependent sample selection is performed. The qualitative figures are intended primarily for scene-level visual comparison, while detailed reconstruction quality is assessed quantitatively using PSNR, SSIM, SAM, and MAE.

*1) Evaluation on the SEN12Ms-CR Dataset:* The SEN12MS-CR dataset covers diverse climate zones and land-cover types and provides a large-scale benchmark for SAR–optical cloud removal. As shown in Table I, HySADNet achieves the best reported values among the compared methods for PSNR, SSIM, SAM, and MAE under the adopted experimental protocol. Compared with the second-best model, HPN-CR, HySADNet improves the PSNR by 0.26 dB and achieves an SSIM of 0.9028. Notably, in terms of spectral fidelity, it achieves the lowest SAM value

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TABLE I
QUANTITATIVE RESULTS ON SEN12MS-CR DATASET

| Method | PSNR | SSIM | SAM | MAE |
|---|---|---|---|---|
| DSen2-CR [23] | 28.5551 | 0.8582 | 9.3443 | 0.0295 |
| USSRN-CR [35] | 29.1851 | 0.8856 | 8.6527 | 0.0271 |
| GLF-CR [24] | 29.5814 | 0.8926 | 8.1947 | 0.0259 |
| TASANet [20] | 29.7105 | 0.8963 | 8.1607 | 0.0260 |
| HS2P [25] | 29.3052 | 0.8994 | 7.7835 | 0.0266 |
| DGMR [21] | 29.9362 | 0.8965 | 7.9816 | 0.0249 |
| HPN-CR [27] | 30.1083 | 0.8951 | 8.0929 | 0.0244 |
| **HySADNet (Our)** | **30.3691** | **0.9028** | **7.6223** | **0.0239** |

TABLE II
QUANTITATIVE RESULTS ON LUOJIASET-OSFCR DATASET

| Method | PSNR | SSIM | SAM | MAE |
|---|---|---|---|---|
| DSen2-CR [23] | 31.7050 | 0.9195 | 3.6958 | 0.0201 |
| USSRN-CR [35] | 32.0890 | 0.9200 | 3.5950 | 0.0200 |
| GLF-CR [24] | 33.8682 | 0.9331 | 2.8612 | 0.0153 |
| TASANet [20] | 34.3965 | 0.9437 | 2.5445 | 0.0144 |
| HS2P [25] | 34.0419 | 0.9347 | 2.7951 | 0.0148 |
| DGMR [21] | 34.4737 | 0.9329 | 2.6263 | 0.0140 |
| HPN-CR [27] | 34.5588 | 0.9452 | 2.5223 | 0.0142 |
| **HySADNet (Our)** | **35.2638** | **0.9479** | **2.2276** | **0.0130** |

TABLE III
SEGMENTATION VALIDATION OF RECONSTRUCTED OPTICAL IMAGES ON THE LUOJIASET-OSFCR DATASET

| Method | Per-class IoU | | | | | | mIoU | F1 |
|---|---|---|---|---|---|---|---|---|
| | Range | Barren | Built-up | Agri. | Water | Forest | | |
| DSen2-CR [23] | 0.4172 | 0.5620 | 0.0773 | 0.2601 | 0.1030 | 0.1656 | 0.2642 | 0.3893 |
| USSRN-CR [35] | 0.5233 | 0.5721 | 0.2983 | 0.3839 | 0.1607 | 0.1977 | 0.3560 | 0.5060 |
| GLF-CR [24] | 0.6295 | 0.5614 | 0.1568 | 0.3541 | 0.1861 | 0.2094 | 0.3496 | 0.4910 |
| TASANet [20] | 0.6141 | 0.6009 | 0.2150 | 0.4341 | 0.2245 | 0.2245 | 0.3815 | 0.5286 |
| HS2P [25] | 0.6269 | 0.5762 | 0.1854 | 0.4036 | 0.1973 | 0.2409 | 0.3717 | 0.5179 |
| DGMR [21] | 0.5716 | 0.6348 | 0.3684 | 0.5329 | 0.2335 | 0.2309 | 0.4287 | 0.5819 |
| HPN-CR [27] | 0.6013 | 0.6265 | 0.3571 | 0.5264 | 0.2290 | 0.2422 | 0.4304 | 0.5833 |
| **HySADNet (Our)** | **0.6348** | **0.6506** | **0.3921** | **0.5463** | **0.2538** | **0.2433** | **0.4535** | **0.5947** |

of 7.6223, demonstrating its superior ability to suppress the spectral shifts commonly observed in cross-modal fusion.

The qualitative results in Fig. 5 provide a scene-level comparison complementary to the quantitative evaluation. In the displayed examples, DSen2-CR exhibits noticeable smoothing in some building edges and road structures, while several other methods show local color or texture inconsistencies in heavily cloud-covered regions. HySADNet generally preserves clearer structural patterns and maintains color appearance closer to the corresponding cloud-free References in these examples.

*2) Evaluation on the LuojiaSET-OSFCR Dataset:* The LuojiaSET-OSFCR dataset contains more fine-grained local textures, placing higher demands on local detail preservation. Table II demonstrates that the performance advantages of HySADNet are further amplified on this dataset. Compared to the advanced Transformer-based methods TASANet and HPN-CR, HySADNet achieves PSNR gains of 0.86 and 0.70 dB over TASANet and HPN-CR, respectively, reaching 35.2638 dB. Furthermore, the MAE is reduced to 0.0130, and the SSIM reaches 0.9479, indicating lower pixel-level reconstruction error under the adopted protocol.

The qualitative comparison in Fig. 6 further illustrates the reconstruction behavior on scenes containing fine-grained spatial structures. In the displayed examples, HySADNet preserves several land-cover boundaries and local spatial patterns that are visually consistent with the corresponding cloud-free references, while some competing methods exhibit local texture or structural inconsistencies in cloud-obscured regions.

## V. ANALYSIS AND DISCUSSION

### A. Analysis Under Different Cloud Coverage

Cloud coverage directly affects the amount of observable information retained in optical imagery and therefore provides a useful condition for examining reconstruction performance under different levels of occlusion. We partition the test samples into five cloud-coverage intervals ranging from 0% to 20% to 80% to 100%. As cloud coverage increases, all compared

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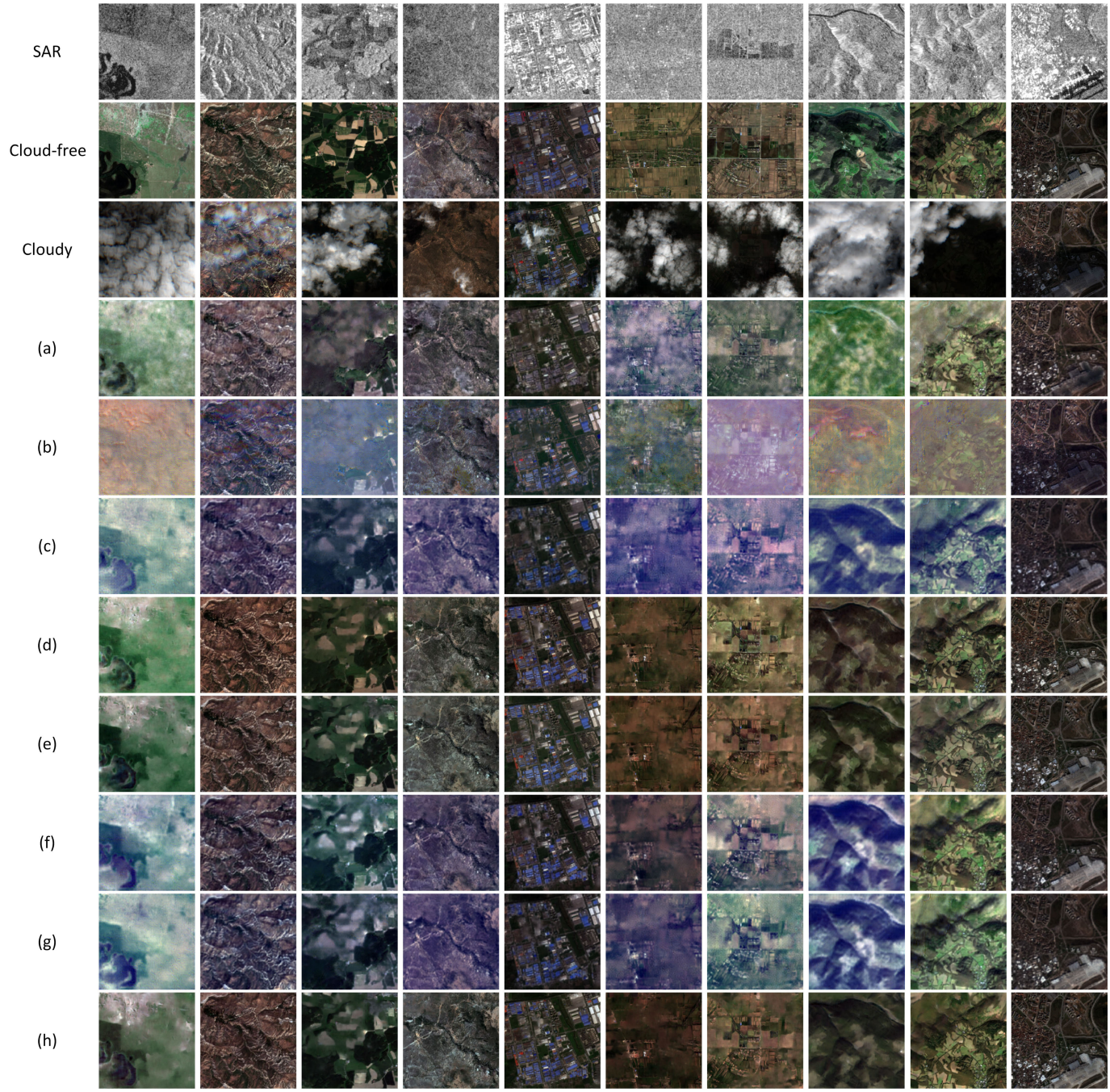



Fig. 5. Qualitative evaluation results on the SEN12MS-CR dataset. From top to bottom: SAR image, cloud-free optical image, cloud-contaminated optical image. (a)–(h) Represent cloud removal results of DSen2-CR, USSRN-CR, GLF-CR, TASANet, HS2P, DGMR, HPN-CR, and our proposed HySADNet model.

methods show a general decline in PSNR and SSIM. HySADNet exhibits a relatively gradual performance decrease across the evaluated intervals. In the 80%–100% interval, DSen2-CR reaches approximately 26 dB PSNR, whereas HySADNet maintains approximately 28.5 dB and also obtains favorable SSIM and MAE values relative to the compared methods.

As shown in Figs. 7 and 8, reconstruction becomes more challenging in the 80%–100% cloud-coverage interval. In the displayed examples, several comparison methods exhibit visible color or texture inconsistencies, whereas the HySADNet results retain relatively consistent scene structures and color appearance with respect to the cloud-free references. These qualitative observations are consistent with the quantitative trends in Fig. 7.

### *B. Analysis Under Different Land-Cover*

High-quality cloud-free images should not only appear visually natural but also serve downstream high-level vision tasks, such as land-cover classification. Semantic consistency is therefore an important metric for evaluating reconstruction quality. We conducted semantic segmentation experiments on the LuojiaSET-OSFCR dataset using a pretrained DeepLabv3

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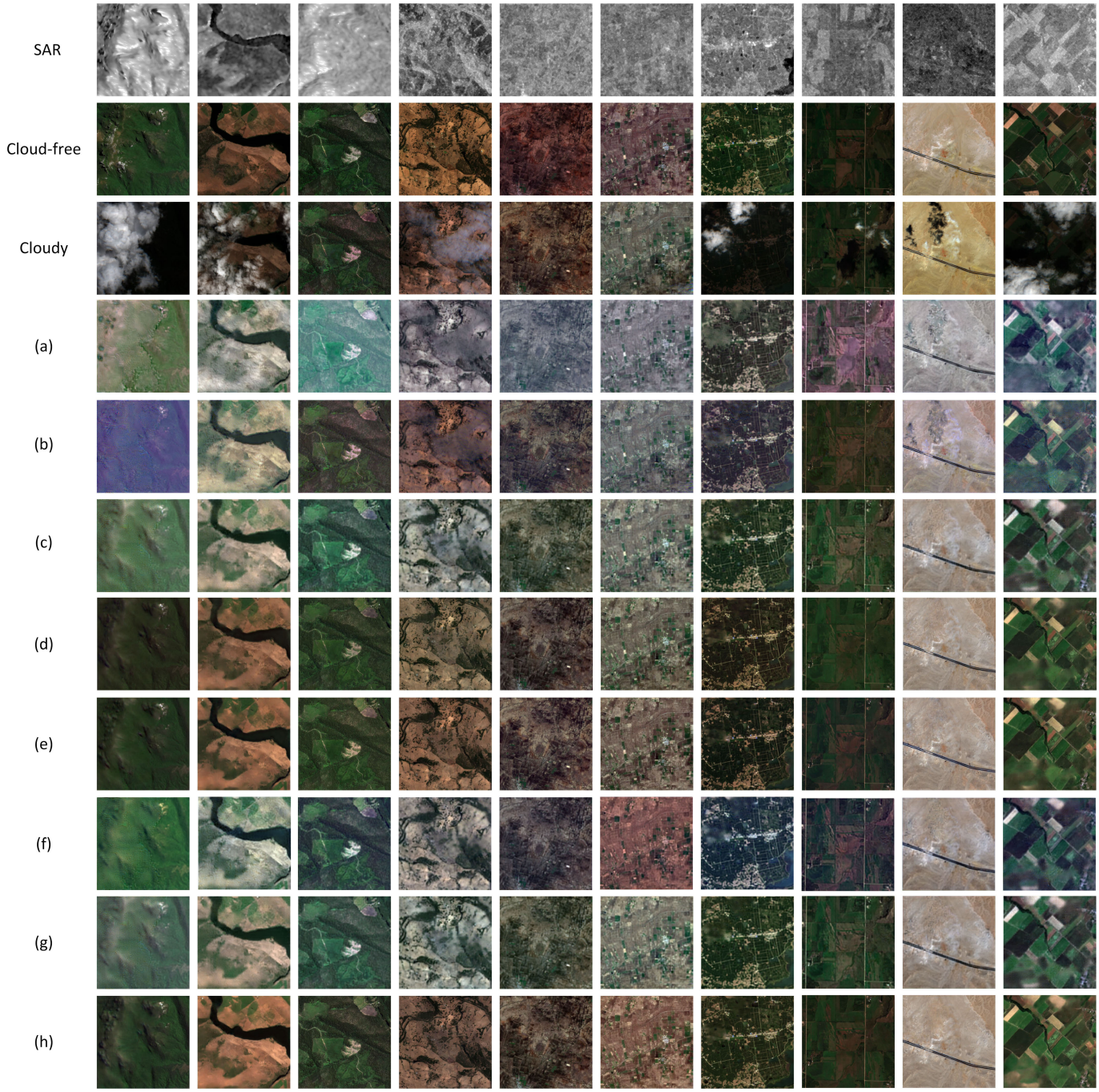



Fig. 6. Qualitative evaluation results on the LuojiaSET-OSFCR dataset. From top to bottom: SAR image, cloud-free optical image, cloud-contaminated optical image. (a)–(h) Represent cloud-removal results of DSen2-CR, USSRN-CR, GLF-CR, TASANet, HS2P, DGMR, HPN-CR, and our proposed HySADNet model.

+ model (trained on cloud-free optical images) on the reconstructed outputs. As shown in Table III and Fig. 9, we evaluated six land-cover categories both qualitatively and quantitatively. HySADNet achieves the highest mean intersection over union (mIoU) of 0.4535 among all compared methods. Among these categories, "Range land" and "Barren land" achieve the highest classification accuracy, likely due to their relatively simple structure. For more structurally complex categories such as "Built-up area" and "Agricultural land," HySADNet attains the highest IoU scores of 0.3921 and 0.5463, respectively. These results indicate that HySADNet preserves useful object-boundary information and shows fewer boundary errors under the evaluated segmentation setting.

### C. Ablation Study and Interpretability Analysis

To verify the necessity and individual contributions of the core components in HySADNet, we conducted stepwise ablation experiments on the LuojiaSET-OSFCR dataset. This section provides a comprehensive analysis ranging from quantitative performance gains to qualitative feature interpretability.

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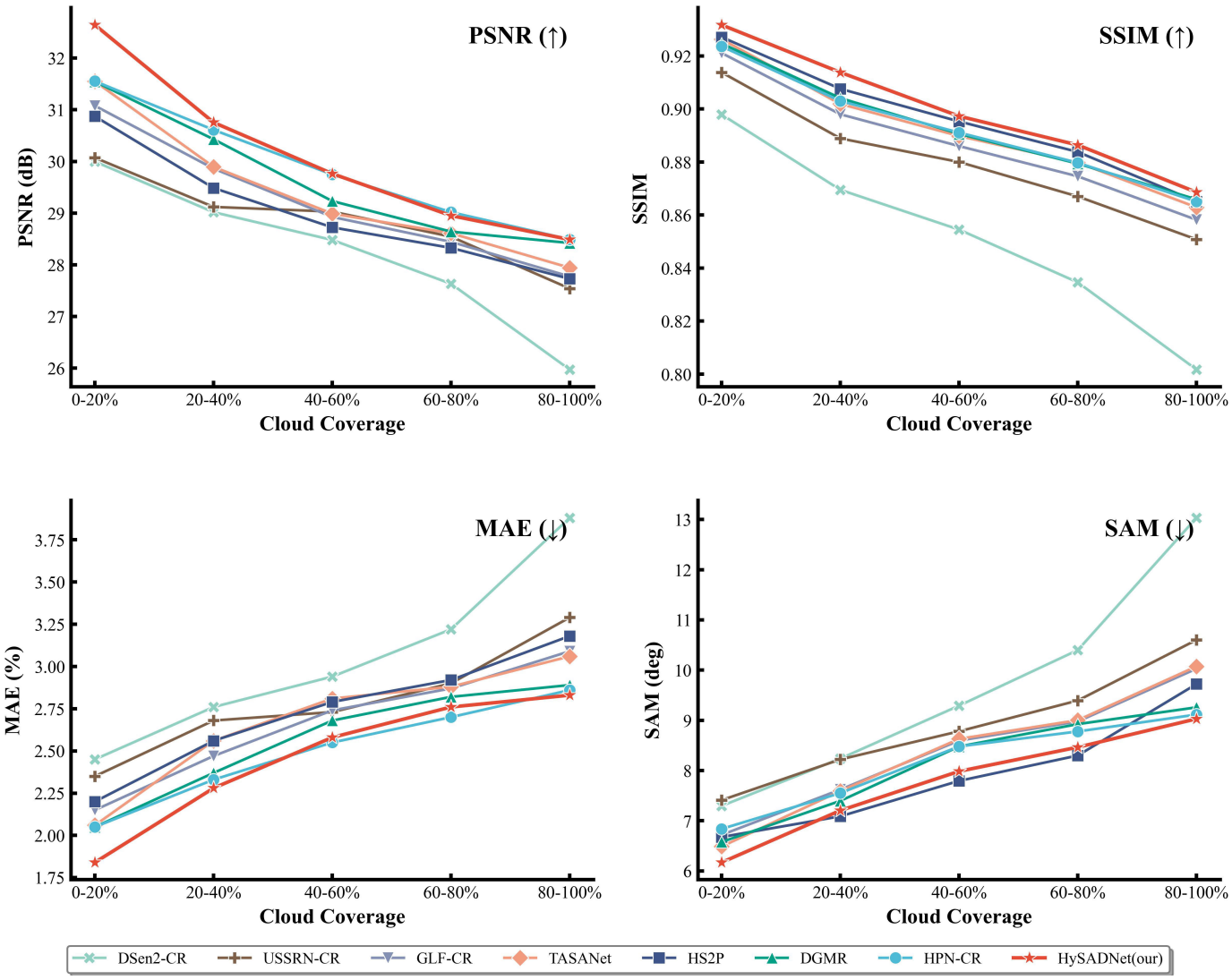



Fig. 7. Line charts of cloud-removal performance metrics under different cloud coverage levels on the SEN12MS-CR dataset.

TABLE IV
ABLATION STUDY OF CORE HYSADNET MODULES

| CSCM | DKF | SAHTB | PSNR | SSIM | SAM | MAE |
|---|---|---|---|---|---|---|
| | | | 32.2649 | 0.9179 | 3.2752 | 0.0227 |
| ✓ | | | 34.1434 | 0.9327 | 2.7472 | 0.0167 |
| | ✓ | | 33.3921 | 0.9243 | 3.0912 | 0.0191 |
| | | ✓ | 33.0438 | 0.9207 | 3.1164 | 0.0203 |
| ✓ | ✓ | | 34.8537 | 0.9402 | 2.3211 | 0.0134 |
| ✓ | | ✓ | 34.1466 | 0.9341 | 2.6328 | 0.0154 |
| | ✓ | ✓ | 33.9848 | 0.9302 | 2.7746 | 0.0170 |
| ✓ | ✓ | ✓ | **35.2638** | **0.9479** | **2.2276** | **0.0130** |

*1) Component-Removal Ablation:* The quantitative results of the ablation study, summarized in Table IV, systematically demonstrate the individual contributions of each core module within the HySADNet framework. Starting from a baseline hierarchical encoder–decoder that achieves a benchmark PSNR of 32.26 dB, the incremental integration of the proposed components leads to consistent performance gains. Notably, the introduction of the CSCM module yields the most substantial performance leap of 1.88 dB, underscoring that cross-modal attention-based calibration is a fundamental prerequisite for bridging the intrinsic modality gap between SAR structural anchors and optical spectral features. Building upon this foundation, the integration of the DKF module further elevates the PSNR to 34.85 dB, validating the effectiveness of pixel-level adaptive multiscale context aggregation in mitigating the information loss typically caused by fixed

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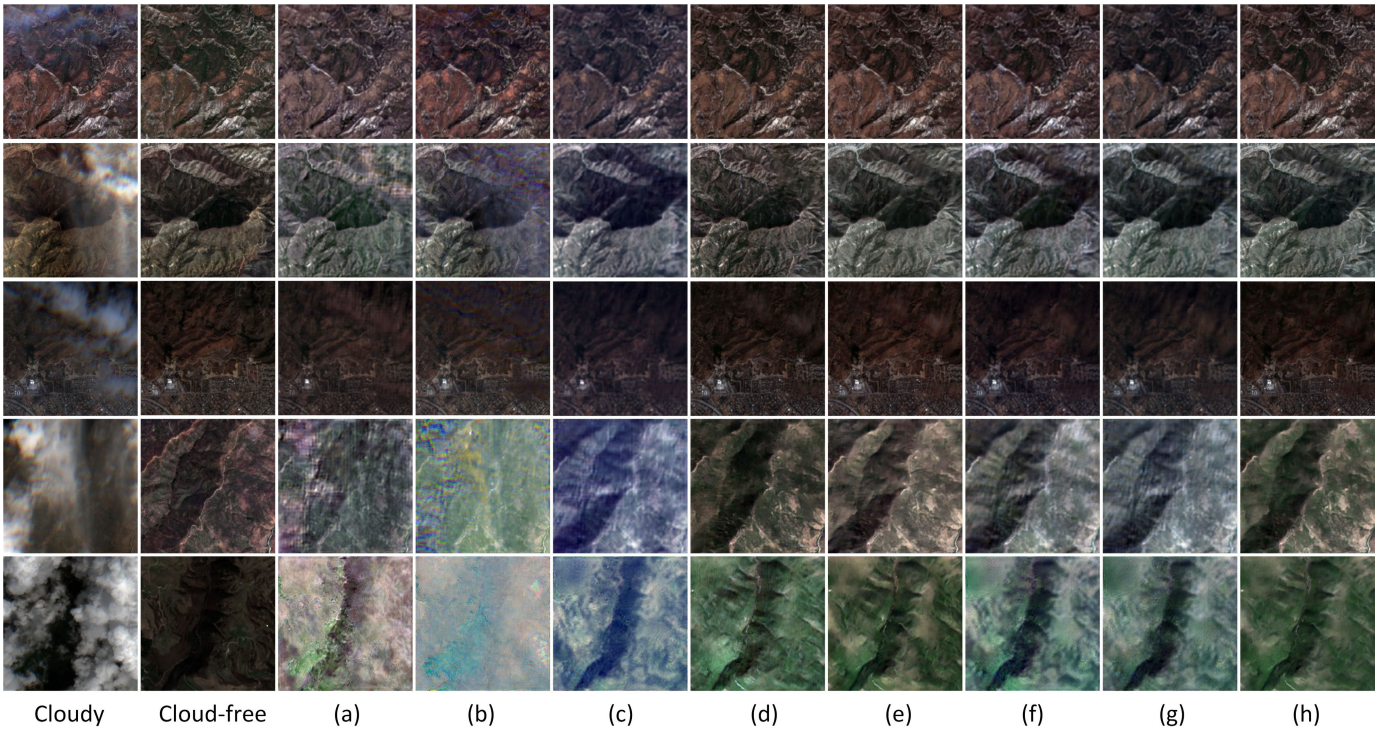



Fig. 8. Qualitative results under different cloud coverage levels on the SEN12MS-CR dataset. (a)–(h) Represent cloud-removal results of DSen2-CR, USSRN-CR, GLF-CR, TASANet, HS2P, DGMR, HPN-CR, and our proposed HySADNet model.

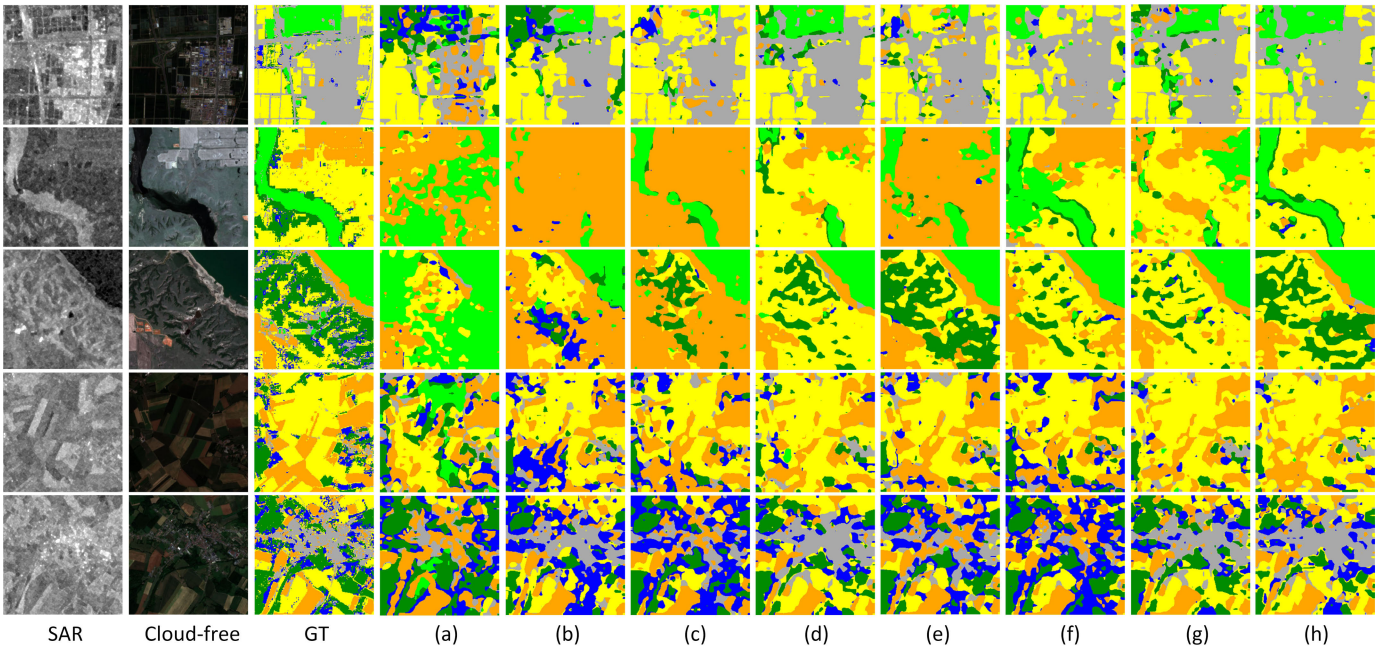



Fig. 9. Comparison of semantic recovery on the LuojiaSET-OSFCR dataset. (a)–(h) Segmentation results obtained from the reconstructed images of DSen2-CR, USSRN-CR, GLF-CR, TASANet, HS2P, DGMR, HPN-CR, and our proposed HySADNet model.

convolutional kernels when dealing with diverse cloud morphologies. Finally, the complete HySADNet configuration reaches its peak performance with a PSNR of 35.26 dB and an SSIM of 0.95, indicating that SAHTB contributes additional reconstruction capability within the complete HySADNet configuration by combining long-range dependence modeling with local-texture representation.

*2) Controlled Comparison of Cross-Modal Fusion Strategies:* As shown in Table V, to determine whether the benefit of CSCM arises from its calibration strategy rather than increased model capacity, we replace CSCM with five representative alternatives while keeping its insertion location, the remaining network structure, and the training protocol unchanged. CSCM achieves the highest PSNR (35.2638 dB), the lowest

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TABLE V
CONTROLLED COMPARISON OF CROSS-MODAL FUSION STRATEGIES

| Fusion strategy | Params (M) | PSNR | SSIM | SAM | MAE |
|---|---|---|---|---|---|
| Concat+Conv | 5.1 | 34.9824 | 0.9471 | 2.3004 | 0.0136 |
| SE | 5.3 | 35.1883 | 0.9478 | 2.2517 | 0.0131 |
| CBAM | 5.3 | 35.1145 | 0.9476 | 2.2719 | 0.0133 |
| FiLM | 5.1 | 34.9209 | 0.9473 | 2.3184 | 0.0138 |
| Cross-attention | 5.2 | 35.1328 | **0.9479** | 2.2663 | 0.0133 |
| **CSCM (ours)** | **5.1** | **35.2638** | **0.9479** | **2.2276** | **0.0130** |

TABLE VI
CONTROLLED COMPARISON OF CONTEXT STRATEGIES

| Strategy | Params (M) | PSNR | SSIM | SAM | MAE |
|---|---|---|---|---|---|
| Single $3 \times 3$ Conv | 5.0 | 35.1224 | 0.9475 | 2.2654 | 0.0133 |
| Dilated Conv | 5.0 | 35.2396 | 0.9467 | 2.2340 | 0.0131 |
| Fixed multi-scale | 5.2 | 35.2207 | 0.9482 | 2.2406 | 0.0131 |
| Selective kernel | 5.2 | 35.1882 | **0.9483** | 2.2489 | 0.0132 |
| **DKF (ours)** | **5.1** | **35.2638** | 0.9479 | **2.2276** | **0.0130** |

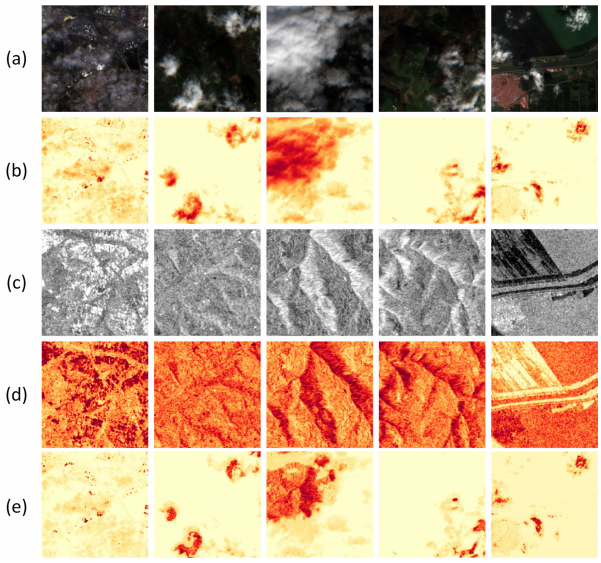



Fig. 10. Cross-modal synergistic calibration and fusion analysis. (a) Cloud-contaminated optical image. (b) Heatmap of optical features. (c) SAR image. (d) Heatmap of SAR features. (e) Heatmap of cross-modal fused features.

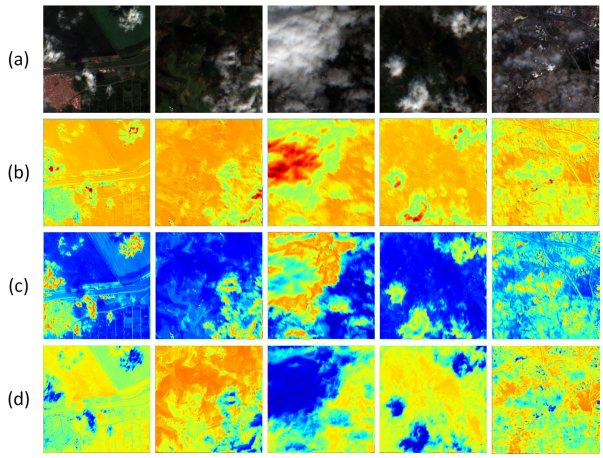



Fig. 11. Multiscale adaptive perception analysis. (a) Cloud-contaminated optical image. (b)–(d) Heatmaps of $3 \times 3$, $5 \times 5$, and $7 \times 7$ convolutional receptive fields.

SAM (2.2276), and the lowest MAE (0.0130) among the six strategies, while also requiring the fewest parameters (5.1 M). Standard cross-attention achieves the highest SSIM of 0.9479, which is equal to CSCM after rounding to four decimal places. Compared with SE, CSCM improves PSNR by 0.0755 dB while reducing the reported parameter count from 5.3 to 5.1 M. Because each configuration is evaluated from a single fixed-seed run, these small numerical differences are interpreted descriptively rather than as evidence of statistical significance.

*3) Controlled Comparison of Context Strategies:* As shown in Table VI, to distinguish adaptive context selection from simple increases in kernel size or model capacity, we replace DKF with four representative alternatives while keeping the remaining network unchanged. DKF achieves the highest PSNR (35.2638 dB), the lowest SAM (2.2276), and the lowest MAE (0.0130). Compared with a single $3 \times 3$ convolution, DKF improves PSNR by 0.1414 dB with only 0.1 M additional parameters. Compared with dilated convolution, DKF improves all four reconstruction metrics, although the PSNR difference of 0.0242 dB is marginal under the single-seed setting. Fixed multiscale and selective-kernel aggregation each contain approximately 0.1 M more parameters than DKF. Selective-kernel aggregation achieves the highest SSIM, exceeding DKF by 0.0004.

*4) Controlled Comparison of Encoder Structures:* As shown in Table VII, the encoder comparison keeps CSCM, DKF, decoder, loss, data, and training settings unchanged. CNN–CNN uses the same 26/52/104-channel CNN hierarchy for both modalities, whereas Transformer–Transformer uses dimensions 26/52/104, depths 8/8/8, heads 2/4/4, and windows 8/12/16 for both streams. The asymmetric design improves PSNR by 1.8888 dB over CNN–CNN while reducing the reported parameter count from 5.8 to 5.1 M. Compared with Transformer–Transformer, its PSNR is only 0.0064 dB lower, while its reported parameter count decreases from 7.3 to 5.1 M and its MAE is lower. We therefore select the asymmetric design for its favorable accuracy–model-size tradeoff rather than claiming that it dominates the fully Transformer-based alternative on every metric.

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TABLE VII
CONTROLLED COMPARISON OF ENCODER STRUCTURES

| Encoder | Params (M) | PSNR | SSIM | SAM | MAE |
|---|---|---|---|---|---|
| CNN–CNN | 5.8 | 33.3750 | 0.9406 | 2.7536 | 0.0168 |
| Transformer–Transformer | 7.3 | **35.2702** | **0.9490** | **2.2180** | 0.0132 |
| **Asymmetric (ours)** | **5.1** | 35.2638 | 0.9479 | 2.2276 | **0.0130** |

TABLE VIII
TEST-TIME SENSITIVITY TO DEGRADED SAR GUIDANCE WITHOUT RETRAINING

| SAR condition | PSNR | SSIM | SAM | MAE |
|---|---|---|---|---|
| Clean (ours) | 35.2638 | 0.9479 | 2.2276 | 0.0130 |
| $\sigma = 0.01$ | 35.2514 | 0.9479 | 2.2308 | 0.0130 |
| $\sigma = 0.03$ | 34.8465 | 0.9464 | 2.3344 | 0.0135 |
| $\sigma = 0.05$ | 33.5135 | 0.9428 | 2.7029 | 0.0153 |
| Missing SAR | 27.3936 | 0.9256 | 5.2032 | 0.0268 |

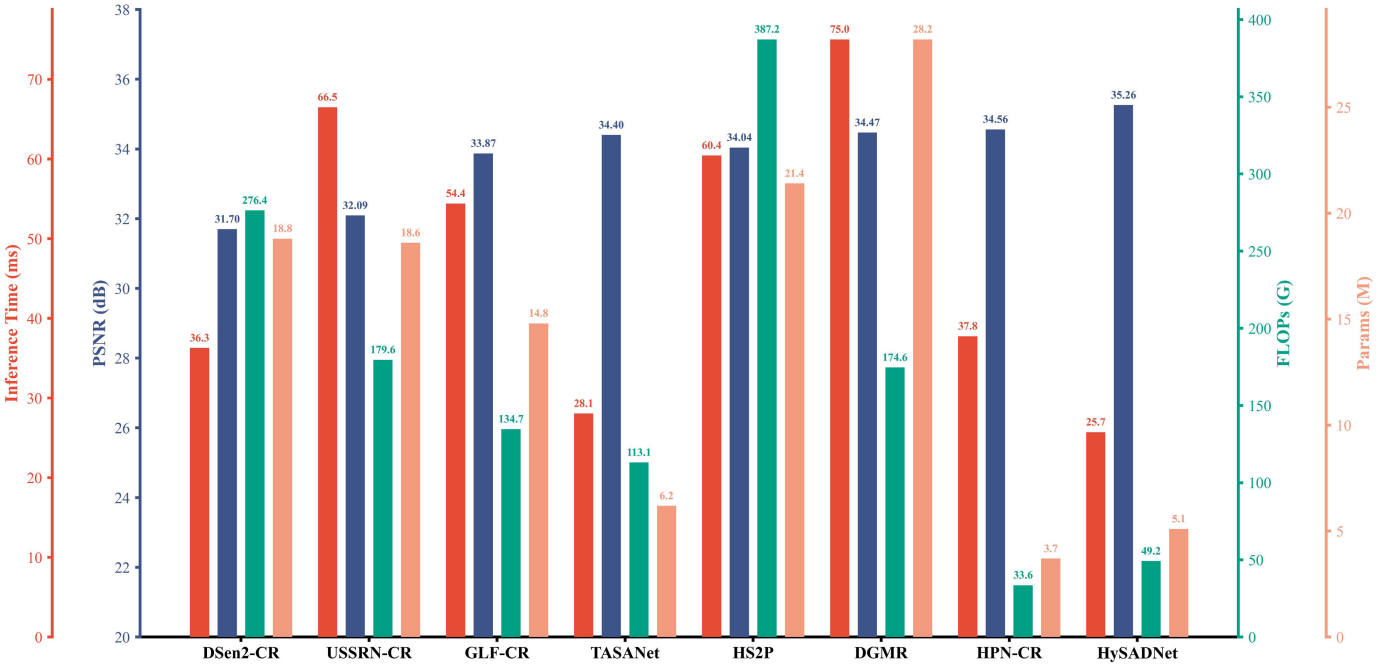



Fig. 12. Comparison of computational efficiency and performance.

*5) Sensitivity to Degraded SAR Guidance:* Since SAR provides auxiliary structural information to HySADNet, we evaluate the sensitivity of the trained model to degraded SAR observations without retraining (Table VIII). The same checkpoint and 4000 evaluation samples are used under five conditions: clean SAR, Gaussian perturbations with $\sigma = 0.01$, 0.03, and 0.05 followed by clipping to $[0, 1]$, and missing SAR represented by an all-zero tensor. Relative to clean SAR, the PSNR loss is 0.0124 dB at $\sigma = 0.01$ and 0.4173 dB at $\sigma = 0.03$. The degradation becomes more pronounced at $\sigma = 0.05$, where PSNR decreases by 1.7503 dB. Removing SAR guidance entirely causes a substantially larger decrease of 7.8702 dB. These results show that the model tolerates mild SAR noise but remains dependent on informative structural guidance under severe degradation.

*6) Interpretability of Cross-Modal Synergistic Calibration:* To further interpret the 1.88-dB gain from CSCM, we visualized the feature interaction process in Fig. 10. In the heatmap visualization, red regions indicate high response values (corresponding to cloud occlusions or significant structural features), while pale yellow indicates low-response background. In the optical feature maps, cloud-affected regions appear as large red blocks, indicating that the original spectral features are severely contaminated by cloud noise and the textures are largely lost. On the contrary, SAR feature maps remain unaffected by clouds and clearly reflect surface structural textures, providing a reliable structural prior for restoration. After passing through CSCM, the fused feature maps exhibit significant interaction and fusion: red high-response cloud regions in the optical features are effectively suppressed into pale yellow backgrounds, while clear SAR structural textures are accurately injected into these regions. This visualization demonstrates that CSCM is content-aware and capable of precisely locating corrupted regions, leveraging SAR structural information to perform guided restoration, achieving deep cross-modal feature complementation.

*7) Analysis of Multiscale Adaptive Perception:* The contribution of the DKF module was further investigated through

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the heatmap analysis of its multibranch kernels (Fig. 11). Different kernel sizes exhibit distinct attention patterns when processing cloudy images: The $3 \times 3$ branch emphasizes fine high-frequency details, focusing on local textures and edges, but is sensitive to noise under large cloud coverage. The $5 \times 5$ branch serves as a transitional scale, balancing local details and regional features. The $7 \times 7$ branch provides broader spatial coverage, effectively perceiving large cloud distributions and macrocontextual information. Experimental results indicate that single-scale kernels cannot simultaneously preserve fine details and handle large-area occlusions. The DKF module dynamically integrates the advantages of these three branches through weighted fusion, significantly enhancing robustness against cloud interference at varying scales.

### D. Computational Efficiency Analysis

Following the profiling protocol described in the implementation details, we evaluate the computational efficiency of HySADNet in terms of parameter count, FLOPs, inference latency, and reconstruction accuracy. As shown in Fig. 12, HySADNet contains 5.1 M parameters and requires 32.185 G FLOPs for a $256 \times 256$ SAR–optical image pair. The measured latency is $42.57 \pm 5.53$ ms with a batch size of 1 under FP32 precision, corresponding to approximately 23.5 image pairs/s.

Compared with heavier Transformer- and diffusion-based models, HySADNet provides a favorable accuracy–complexity tradeoff. Although HPN-CR has a smaller computational footprint, HySADNet achieves higher reconstruction accuracy with a moderate increase in computational cost. Since inference speed depends on hardware and implementation details, we report the measured latency under the specified setting rather than making an unconditional real-time-processing claim.

## VI. Conclusion

In this study, we presented HySADNet, an asymmetric SAR-guided network for paired SAR–optical cloud removal. The framework assigns different representation roles to the optical reconstruction and SAR guidance streams, employs reciprocal cross-modal feature calibration to integrate heterogeneous information, and uses adaptive multiscale context aggregation together with global–local representation learning for cloud-contaminated optical reconstruction. Experiments on SEN12MS-CR and LuojiaSET-OSFCR show competitive reconstruction performance in terms of spatial and spectral quality, while the downstream semantic-segmentation evaluation indicates that the reconstructed imagery preserves useful information for the evaluated high-level vision task. Controlled replacement experiments further characterize the reconstruction–complexity tradeoffs of the fusion, context, and encoder designs.

However, as HySADNet relies on paired and spatially registered SAR–optical observation data, its response to geometric registration errors remains to be further validated. The evaluation data are limited to the SEN12MS-CR and LuojiaSET-OSFCR datasets; therefore, the reported performance alone does not demonstrate its generalization capability across different geographical regions, seasons, sensors, or datasets. Controlled architecture comparisons were conducted using a fixed random seed for each comparison group; consequently, minor performance differences should be regarded as empirical observations under the current training protocol rather than statistically significant evidence. SAR degradation experiments further indicate that, although the network is relatively insensitive to minor perturbations, reconstruction quality declines markedly when the SAR input is severely degraded; in the absence of SAR guidance, reconstruction quality deteriorates significantly. These observations suggest that HySADNet is currently best suited to paired SAR–optical cloud-removal scenarios, where the two modalities can be reliably registered, and the auxiliary SAR observations retain sufficient structural information.

Future work will investigate registration-tolerant and reliability-aware cross-modal fusion, region-disjoint and cross-dataset evaluation, transfer across sensors and seasonal conditions, and repeated-seed uncertainty analysis. These directions will provide a more complete assessment of the generalization and operational robustness of SAR-assisted optical cloud-removal models.

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